OFFIONG BASSEY EDET

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RESEARCH INTERESTS

Low-Resource Natural Language Processing (NLP), Machine Translation, Information Extraction, Multilingual Language Technologies, Speech Synthesis (TTS/ASR) and Deep Learning.

EDUCATION

University of Cross River State | 2019 – 2023
B.Sc. Computer Science - Second Class Upper

RESEARCH EXPERIENCE & PROJECTS

Efik Machine Translation System for a Low-Resource Language | Present

- Built a high-quality MT system for Efik, a low-resource African language.
- Curated and cleaned a parallel corpus of 13,865 English–Efik sentences in collaboration with linguists and native speakers.
- Fine-tuned mT5-base using PyTorch and Hugging Face Transformers.
- Achieved BLEU 15.61 and chrF 35.04, validated with human evaluation.
- Paper submitted to AfricaNLP at EACL 2026, detailing dataset and methodology.
- **Tools:** Python, PyTorch, HuggingFace, mT5

Efik Text-to-Speech (TTS) using VITS & Diffusion Models | Present

- Developing a natural-sounding TTS system for Efik, aligned with MT work.
- Preprocessing and aligning ~10 hours of raw audio data.
- Experimenting with **VITS** and diffusion-based TTS architectures for sample quality and efficiency.
- Conducting evaluation using **objective metrics and human listening tests** to validate naturalness and intelligibility.
- Paper to be submitted to Interspeech 2026.
- **Tools:** Python, PyTorch, Audacity, Google Colab

Life Jacket Detection System | *2025*

- Developed a real-time object detection system using a fine-tuned YOLOv5 model and OpenCV to enhance maritime safety.
- **Tools:** Python, YOLO, OpenCV, TensorFlow

PUBLICATIONS & PREPRINTS

Edet, O., et al. (2025). "**Developing an English-Efik Corpus and Machine Translation System for Digitization Inclusion.**" Submitted to **AfricaNLP** at **EACL 2026**.
(Preprint available upon request)

RESEARCH EXPERIENCE

Masakhane NLP Community |*Contributor* | (2025)

- Built low-resource language datasets and ML pipelines for translation and TTS.

Bouquet AI Research by Meta |*Efik Translator* | (2025)

- Contributed to building a machine translation corpus in collaboration with linguists and native speakers.

ML Collective | *Contributor* | (2025)

- Assisted in community projects applying ML for social good.

Data Science Nigeria (DSN) | *AI & ML Member* | (2025)

- Participated in AI challenges, bootcamps**,** and nationwide skill-building programs.

PROFESSIONAL EXPERIENCE

Plotweaver | Machine Learning Engineer – NLP | Jan 2026 – Present

- Designed and fine-tuned transformer-based models for low-resource language processing, including speech recognition and text generation tasks.
- Built end-to-end NLP pipelines for data collection, preprocessing, augmentation, and evaluation across multilingual datasets.
- Improved model performance through dataset curation, error analysis, and iterative fine-tuning.
- Deployed scalable inference pipelines using Python and cloud infrastructure for real-time and batch processing workloads.

RapidMind AI (Remote) | *AI Engineer* | Jun 2025 – Oct 2025

- Developed Rasa NLU chatbot for fintech customer support automation.

Klas (Remote, Toronto) | *Full Stack Engineer* | May 2024 – Present

- Implemented and optimized cloud-based infrastructure for ML-enabled applications.
- Improved backend performance and scalability by optimizing server architecture and backend services using Nodejs and Typescript, leading to reduced latency and better traffic management.
- Leveraged AWS Services, including, EC2 Instances, Lambda functions, Dynamo DB, and S3, to optimize image processing, file uploads, and other cloud-based functionalities, contributing to a more efficient and scalable infrastructure.

Sycamore NG - *Financial Services Company* | Backend Developer | *June 2023 – October 2023*
Lagos, GRA, Hybrid

- Utilized backend engineering expertise to write efficient Nodejs REST APIs and conducted thorough testing to ensure the application met requirements and was free from bugs.
- Conducted rigorous research and provided extensive reports on products.
- Normalized the database to reduce latency and redundancy, and to increase performance.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL
ML/AI: PyTorch, TensorFlow, HuggingFace Transformers, Scikit-learn, FastSpeech2, VITS, mT5, CNNs, LLM fine-tuning, CrewAI
Cloud & Tools: AWS (EC2, Lambda, S3, DynamoDB), Git/GitHub, Docker, Postman
Other Frameworks: React.js, Node.js, Express.js

CERTIFICATIONS

- **Deep Learning Specialization**, DeepLearning.AI | 2025
- **Machine Learning**, IBM | 2023
- **Introduction to Software Engineering**, IBM | 2023